Lab Exercise 8- Create POD in Kubernetes

Objective:

- Understand the basic structure and syntax of a Kubernetes Pod definition file (YAML).
- Learn to create, inspect, and delete a Pod in a Kubernetes cluster.

Prerequisites

- Kubernetes Cluster: You need a running Kubernetes cluster. You can set up a local cluster using tools like Minikube or kind, or use a cloud-based Kubernetes service.
- kubectl: Install and configure kubectl to interact with your Kubernetes cluster.
- Basic Knowledge of YAML: Familiarity with YAML format will be helpful as Kubernetes resource definitions are written in YAML.

Step-by-Step Guide

Step 1: Create a YAML File for the Pod

We'll create a Pod configuration file named **pod-example.yaml**

```
apiVersion: v1
kind: Pod
metadata:
  name: my-pod
  labels:
    app: web
spec:
  containers:
    - name: my-container
      image: nginx:latest
```

Explanation of the YAML File

- apiVersion: Specifies the version of the Kubernetes API to use. For Pods, it's typically v1.
- kind: The type of object being created. Here it's a Pod.
- metadata: Provides metadata about the object, including name and labels. The name must be unique within the namespace, and labels help in identifying and organizing Pods.
- spec: Contains the specifications of the Pod, including:
    - containers: Lists all containers that will run inside the Pod. Each container needs:
        - name: A unique name within the Pod.
        - image: The Docker image to use for the container.
        - ports: The ports that this container exposes.
        - env: Environment variables passed to the container.

Step 2: Apply the YAML File to Create the Pod

Use the kubectl apply command to create the Pod based on the YAML configuration file.

```
kubectl apply -f pod-example.yaml
```

This command tells Kubernetes to create a Pod as specified in the pod-example.yaml file.

Step 3: Verify the Pod Creation

To check the status of the Pod and ensure it's running, use:

```
kubectl get pods
```

This command lists all the Pods in the current namespace, showing their status, restart count, and other details.

You can get detailed information about the Pod using:

```
kubectl describe pod my-pod
```

This command provides detailed information about the Pod, including its events, container specifications, and resource usage.

Step 4: Interact with the Pod

You can interact with the running Pod in various ways, such as accessing the logs or executing commands inside the container.

View Logs: To view the logs of the container in the Pod:

```
kubectl logs my-pod
```

Execute a Command: To run a command inside the container:

```
kubectl exec -it my-pod -- /bin/bash
```

The -it flag opens an interactive terminal session inside the container, allowing you to run commands.

Step 5: Delete the Pod

To clean up and remove the Pod when you're done, use the following command:

```
kubectl delete pod my-pod
```

This command deletes the specified Pod from the cluster.

```
C:\Users\NK>(echo apiVersion: v1 & echo kind: Pod & echo metadata: & echo   name: my-pod & echo   labels: & echo     app
: web & echo spec: & echo   containers: & echo     - name: my-container & echo       image: nginx:latest) > pod-example.
yaml

C:\Users\NK>type pod-example.yaml
apiVersion: v1
kind: Pod
metadata:
  name: my-pod
  labels:
    app: web
spec:
  containers:
    - name: my-container
      image: nginx:latest

C:\Users\NK>minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
```

```
C:\Users\NK>kubectl apply -f pod-example.yaml
pod/my-pod created

C:\Users\NK>kubectl get pods
NAME     READY   STATUS              RESTARTS   AGE
my-pod   0/1     ContainerCreating   0          3s

C:\Users\NK>kubectl describe pod my-pod
Name:             my-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Tue, 28 Apr 2026 08:46:20 +0530
Labels:           app=web
Annotations:      <none>
Status:           Pending
IP:
IPs:              <none>
Containers:
  my-container:
    Container ID:
    Image:          nginx:latest
    Image ID:
    Port:           <none>
    Host Port:      <none>
    State:          Waiting
      Reason:       ContainerCreating
    Ready:          False
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-tvsbm (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   False
  Initialized                 True
  Ready                       False
  ContainersReady             False
  PodScheduled                True
Volumes:
```

```
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-tvsbm (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   False
  Initialized                 True
  Ready                       False
  ContainersReady             False
  PodScheduled                True
Volumes:
  kube-api-access-tvsbm:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  7s    default-scheduler  Successfully assigned default/my-pod to minikube
  Normal  Pulling    6s    kubelet            Pulling image "nginx:latest"

C:\Users\NK>kubectl logs my-pod
Error from server (BadRequest): container "my-container" in pod "my-pod" is waiting to start: ContainerCreating

C:\Users\NK>kubectl exec -it my-pod -- /bin/bash
```